- **Title:** To acquire, and process EMG signals from a provided device for analytical purposes and so that they may be interpreted medically.

- **Aims:** The aim of this project is to investigate how physiological signals can be used to understand human physiology. We will decide between using an ECG or EMG-based acquisition system, which we will then use to collect data for analysis to create a final presentation to communicate our findings. Alongside achieving the technical objective, the project aims to develop key engineering skills such as teamwork, leadership, project planning and the ability to communicate our findings to a non-technical audience.

- **Scope:** The entire project will be complete by the end of Project Week 1 with a Quality Management Audit on Thursday morning, and individual video submissions outlining how we used biosignals to answer an engineering question due by Thursday afternoon. We will use either an ECG or EMG system to gather data depending on availability and then MATLAB to process and analyse our data. The project is limited to investigating the use of biosignal data to answer an engineering question and will not be used to make any medical judgements or begin considering development of new systems.

- **Objective 1:**
    - Specific - Obtain raw data from the EMG
    - Measurable – Take at least five readings
    - Achievable - Using existing EMG device and test subject
    - Relevant - Required for estimation of contraction duration and muscle activity
    - Time-bound - by 3pm Tuesday

- **Objective 2:**
    - Specific - Use software to detect the features of the EMG signal such as amplitude and frequency range
    - Measurable - Numerical values for RMS, MAV, and mean frequency
    - Achievable - Using MATLAB, which we have access to
    - Relevant – Be able to classify activity and identify muscle weakness
    - Time-bound - To be completed by Thursday midday

- **Objective 3:**
    - Specific - Formulate a presentation to a non-scientific audience
    - Measurable - Timing to 3 minutes of content including 1 graph explained in plain language
    - Achievable - Using Microsoft PowerPoint
    - Relevant - We will be presenting our group's work at the end of the week
    - Time-Bound - To be completed by 10am Friday

- **Objective 4:**

- Specific - Establish and maintain a Quality Management System
- Measurable – Successful audit would mean that we have successfully maintained our system
- Achievable – Have all completed the Quality Management knowledge quiz on Moodle so have knowledge on how to maintain a QMS throughout a project
- Relevant – We will have an audit towards the end of the week
- Time-Bound – Continuous objective with a final assessment on Thursday afternoon

- **Work Package 1: Data Collection**
  - Summary: Collect raw data from the EMG
  - Task 1.1: Prepare the equipment needed
    - Check the equipment works including the electrodes, mark where the electrodes will be placed on the bicep of the test subject
    - *Duration: 20 minutes Dependency: None Day: Tues*
  - Task 1.2: Record readings
    - Take at least five readings, saving and naming each reading into a file
    - *Duration: 45 minutes Dependency: T1.1 and T4.2 Day: Tues*
  - Task 1.3: Check data quality
    - Review data and check for any noise or failed recordings, if so, repeat the recording until we have five usable readings
    - *Duration: 30-60 minutes Dependency: T1.2 Day: Tues*

- **Work Package 2: Signal Analysis**
  - Summary: Process the recordings and extract the features needed to be able to classify activity and identify muscle weakness
  - Task 2.1: Process data in MATLAB
    - Import recordings to MATLAB and calculate RMS, MAV, and mean frequency, saving results to a table
    - *Duration: 60-120 minutes Dependency: T1.3 Day: Tues-Wed*
  - Task 2.2: Interpret the results for activity and muscle weakness
    - Set RMS threshold to separate rest from contraction and calculate contraction duration. Compare RMS, MAV, and mean frequency across the five readings to see how muscle activity changes and if a downward shift is present which may indicate muscle fatigue or weakness. Plot trend graphs, producing the plain-language graph
    - *Duration: 60 minutes Dependency: T2.1 Day: Tues-Wed*

- **Work Package 3: Presentation**
  - Summary: Create a presentation about the group's work to communicate towards a non-scientific audience
  - Task 3.1: Create the presentation

        - Make the slides clear and easy to interpret, include graphs made in T2.3 and data obtained from the EMG and explain them in plain language
        - *Duration: 120 minutes Dependency: T2.3 Day: Thurs*
    - Task 3.2: Rehearse the presentation
        - Practise going through the slides and presenting it to a non-scientific audience, engaging with them and maintaining eye-contact, as well as speaking confidently and aiming for no filler words, timing to 3 minutes
        - *Duration: 60 minutes Dependency: T3.1 Day: Thurs-Fri*

- **Work Package 4: QMS**
    - Summary: Set up QMS folder at the start of the project and continuously keep it up to date throughout the week, checking it before the audit
    - Task 4.1: Set up folder tree
        - Use folder tree similar to an ISO 13485 QMS, provided by the lab script, including folders such as Quality control, Engineering, Operations, Purchasing, Sales/customer service, etc
        - *Duration: 15-30 minutes Dependency: none Day: Mon*
    - Task 4.2: Write file naming procedure
        - Agree with the group and document a file naming convention, saving it under the procedure file in the QMS
        - *Duration: 60 minutes Dependency: T4.1 Day: Mon*
    - Task 4.3: File documents as they are produced
        - Name each file created using the file naming procedure and put into correct folders
        - *Duration: 10 minutes/day Dependency: T4.2 Day: Mon-Fri*
    - Task 4.4: Create documents gallery
        - Create a table in excel for all folders and files, placing in the Documents Gallery section in the QMS, final update after the last document is produced
        - *Duration: 60 minutes Dependency: T4.3 Day: Mon-Thurs*
    - Task 4.5: Complete pre-audit check
        - Go through every folder against the brief, checking if file names follow the procedure, and update misnamed files and fixing any gaps
        - *Duration: 45 minutes Dependency: T4.4 Day: Thurs*

Milestone	**Description**	**RelevantWork package**	**Target time**
M1	QMS folder and file structure established	WP4	Monday afternoon
M2	Partner team selected and data acquisition	WP1	Monday afternoon

	method agreed (EMG preferred over ECG)		
M3	Kit collected and successfully configured	WP1	Tuesday morning
M4	Five sets of data recorded	WP1	Tuesday afternoon
M5	Data features extracted using MATLAB	WP2	Wednesday afternoon
M6	Signal analysis complete and conclusions made	WP2	Thursday midday
M7	QMS ready for auditing	WP4	Thursday morning
M8	Presentations and recordings complete and ready for submission	WP3	Thursday afternoon

- **Deliverables**
  - WP1:
    - D1.1 - Data acquisition setup complete
    - D1.2 - Obtained 5 set of raw data readings
    - D1.3 - Organised dataset stored inside the QMS
  - WP2:
    - D2.1 - MATLAB script complete
    - D2.2 - Feature extraction results
    - D2.3 - Processed graphs from MATLAB
    - D2.4 - Summary and conclusions drawn from readings and graphs
  - WP3:
    - D3.1 - Complete PowerPoint presentation
    - D3.2 - Script/ notes to go alongside presentation
    - D3.3 - Individually submit 3-minute video of the presentation
  - WP4:
    - D4.1 - QMS file naming and folder structure decided
    - D4.2 - Document produced containing both file naming and folder structure
    - D4.3 - Documents gallery created
    - D4.4 - Successful completed audit